Statistical Learning and Machine Learning

Exam Topics - May 2024

TOPIC 1

You are provided with the data points in 2D space, $\{x_i\}_{i=1}^5$, as shown in Figure 1. The same data is also given in the matrix form here:

$$\mathbf{X} = \begin{bmatrix} 1.27 & 1.82 & 1.97 & 2.29 & 2.35 \\ 0.19 & 3.85 & 0.43 & 0.99 & 3.01 \end{bmatrix} \tag{1}$$

Assume the data points are independent and identically distributed according to the multivariate **Gaussian** distribution. In the following, you must estimate the parameters of the multivariate Gaussian distribution from the above data points, using the maximum likelihood (ML) estimation approach.

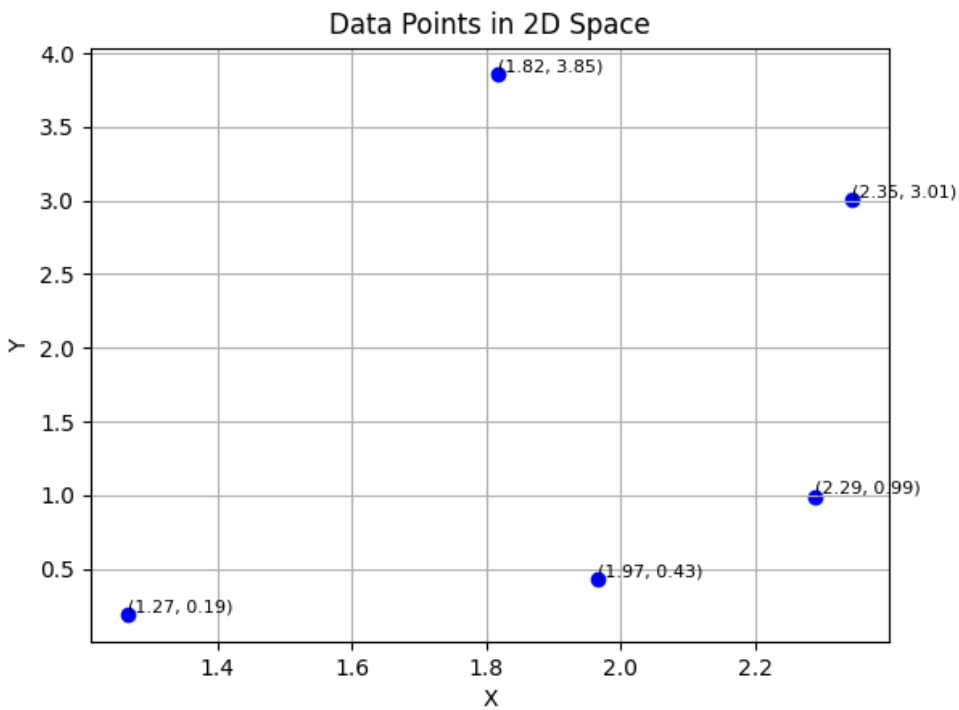

Fig. 1. Data points $\{\mathbf{x}_i\}_{i=1}^5$ residing in the 2D space obtained from a multivariate Gaussian distribution with unknown mean μ and covariance Σ.

1) Estimate the mean of the Gaussian distribution, μ_{ML}, from the available data points by using the maximum likelihood (ML) method. (Score: 25%)
2) Comment on whether the above mean estimate estimate, μ_{ML}, is biased OR unbiased. (Score: 15%)
3) Estimate the covariance matrix of the Gaussian distribution. Σ_{ML}, from the available data points by using the maximum likelihood (ML) method. (Score: 30%)
4) Comment on whether the above covariance matrix estimate, Σ_{ML}, is biased OR unbiased. If the above estimate is biased, how would you make it unbiased? (Score: 30%)

TOPIC 2

You are provided with the data points in 2D space, $\{x_i\}_{i=1}^{5}$, as shown in Figure 2. The same data is also given in the matrix form:

$$\mathbf{X} = \begin{bmatrix} -1 & -0.5 & 0 & 0.5 & 0.75 \\ 0 & 0.8 & 0.5 & 1.25 & 0.75 \end{bmatrix} \tag{2}$$

The following questions relate to the process of applying the principal component analysis (PCA) method on the given data:

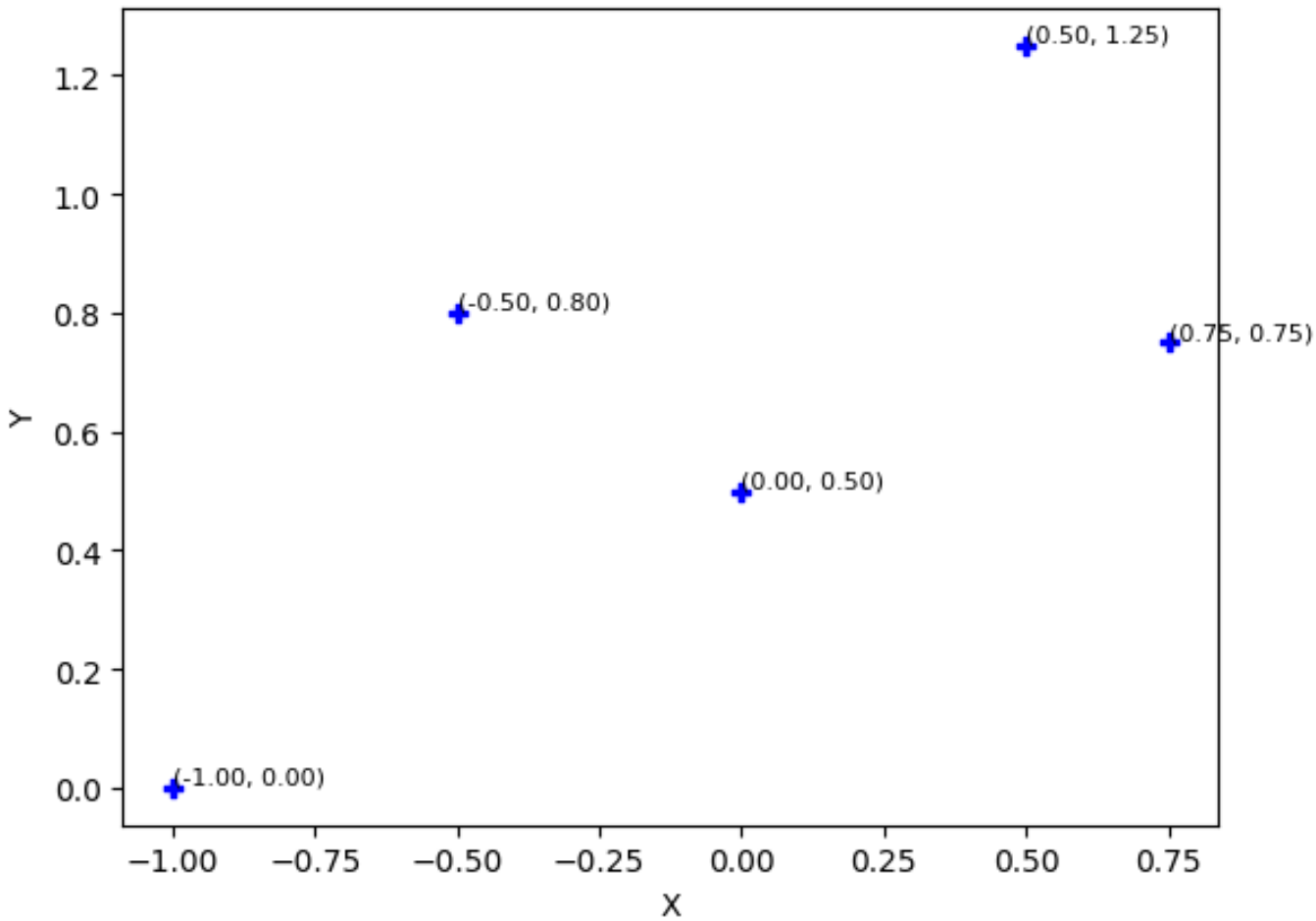

Fig. 2. The input data points residing in the 2D space along with their coordinates.

1) (Score: 30%) Using the PCA method on the data set shown in Figure 2, obtain the *principal components* of the data.
2) (Score: 20%) What are the variances of the projected data along the principal components obtained in the previous step?
3) (Score: 25%) Plot the principal components (corresponding vectors) along with the centered input data.
4) (Score: 25%) You wish to project input data along one of the principal components obtained in the previous step. Which principal component would you choose and why?

TOPIC 3

Consider a data set, represented by $\mathbf{X}$, containing $N = 6$ *training* data points shown by blue-square and red-circle markers in Figure 3.

$$\mathbf{X} = \begin{bmatrix} 2 & 3 & 4 & 6 & 7 & 8 \\ 3 & 4 & 2 & 6 & 5 & 7 \end{bmatrix} \tag{3}$$

The blue-squared data points belong to class $\mathcal{C}_1$, with the corresponding target value $t = 1$, whereas the red-circled points belong to $\mathcal{C}_2$, with the target values $t = 0$. The corresponding target vector becomes $\mathbf{t} = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \end{bmatrix}$. Using this training data, you are required to classify *test data points*, T1=$\begin{bmatrix} 5 \\ 4 \end{bmatrix}$ and T2=$\begin{bmatrix} 6 \\ 4.5 \end{bmatrix}$, that are shown as green triangles in Figure 3, to either $\mathcal{C}_1$ or $\mathcal{C}_2$, by using the *probabilistic generative classification model*.

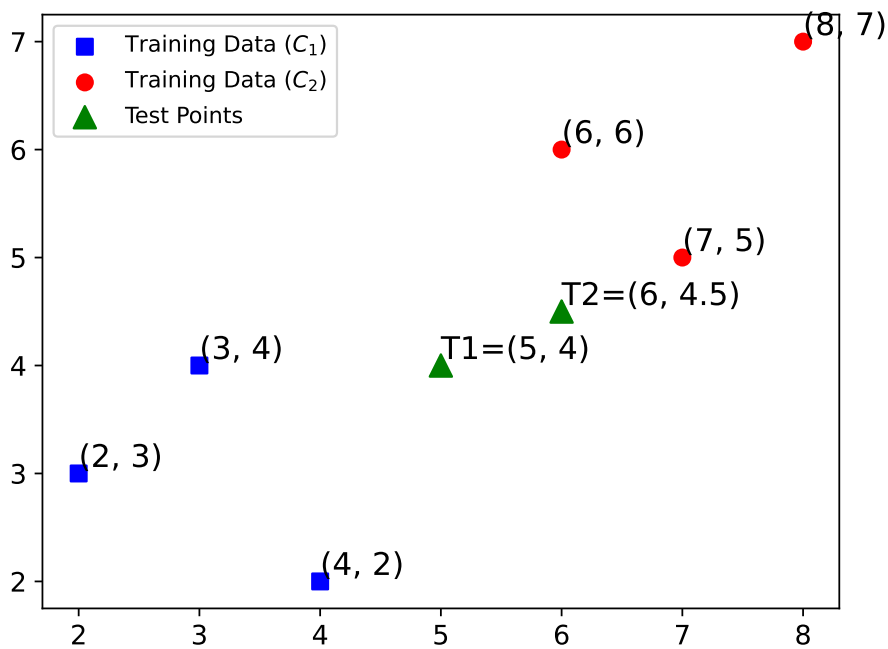

Fig. 3. The graphical illustration of the data set (both training and test sets) used in Topic 2.

1) (Score: 60 %) To apply the *probabilistic generative model* to classify the two test data points to either $\mathcal{C}_1$ or $\mathcal{C}_2$, assume that the class conditional probability densities, i.e., $p(\mathbf{x}|\mathcal{C}_1)$ and $p(\mathbf{x}|\mathcal{C}_2)$ are Gaussian with means $\boldsymbol{\mu}_1$ and $\boldsymbol{\mu}_2$ respectively and the *same* covariance matrix $\boldsymbol{\Sigma}$. Let $p(\mathcal{C}_1)$ and $p(\mathcal{C}_2)$ denote the class priors for the classes $\mathcal{C}_1$ and $\mathcal{C}_2$, respectively. Using the training data points (blue-squares and red circles in Figure 3) and the maximum likelihood approach, calculate the values of the parameters $\boldsymbol{\mu}_1$, $\boldsymbol{\mu}_2$, $\boldsymbol{\Sigma}$, $p(\mathcal{C}_1)$, and $p(\mathcal{C}_2)$. Using these estimated parameter values, compute the posterior probability function $p(\mathcal{C}_1|\mathbf{x})$.
2) (Score: 40 %) In continuation to the above task 1, calculate and plot the *decision boundary* of the probabilistic generative classifier, together with the available data points in Figure 3. Using this decision boundary, which classes will you assign to the test data points T1 and T2?

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TOPIC 4

Consider building and training a feedforward artificial neural network for a regression task, where you are provided with an 8-dimensional input vector, $\mathbf{x_i} \in \mathbb{R}^8, i = 1, \ldots, N$ and the target is a 2-dimensional vector, $\mathbf{t_i} \in \mathbb{R}^2, i = 1, \ldots, N$. Answer the following questions:

1) (Score: 20 %) *Network Construction*: Construct a feed forward artificial neural network with 2 hidden layers, each containing 4 neurons. Draw the complete neural network architecture. Mention the correct dimensions of all weight matrices in each layer.
2) (Score: 20 %) *Parameters and Hyperparameters*: List and describe the parameters and hyperparameters for this network. Provide appropriate symbols or functions along with their correct dimensions.
3) (Score: 20 %) *Activation Functions*: Which activation functions were used in your constructed neural network above, at each layer? Describe the reasons for choosing those specific activation functions for each layer.
4) (Score: 20 %) *Loss Function*: Which *loss function* would you choose for your network, and why? Additionally, discuss whether you would employ the same loss function if your task shifted from regression to classification. Provide reasoning for your choice(s).
5) (Score: 20 %) *Training and hyperparameter selection*: Briefly describe the process to estimate the values of the neural network *parameters* and the *hyperparameters*.